

Issues in History and Philosophy of Science

Sample Exam

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Instructions

Grading: There are 21 questions on the exam. Each question is worth 5 points, for a maximum score of 100. (So the first mistake does not deduct points)

Formula sheet: Attached for your convenience

Exam duration: 90 minutes

Allowed reference materials: Dictionary

The exam is strictly confidential. Exam papers and all related materials must not be removed from the classroom. Please return all documents to the invigilators at the end of the session.

1. Adlerian psychology can explain why a person would risk their life to save a drowning child, and also why the same person would remain passive when seeing the drowning child. According to Popper, what does this not indicate?

- a. That supporters of Adler's theory can provide many inductive arguments in its favor
- b. The strength of Adler's theory, which is able to explain a wide range of possible scenarios
- c. The weakness of Adler's theory, which cannot be refuted
- d. That Adler's theory is pseudo-science

2. Does the (positivist) scientific method allow us to confirm the claim that all people in the world will be 10 centimeters taller tomorrow morning?

- a. No. Because this claim contradicts what is known to science
- b. Yes. Due to the New Riddle of Induction

- c. No. Because confirmation is done by induction, and since all people to date have not grown 10 centimeters overnight, the same will be true tomorrow
- d. Yes. Through the Duhem-Quine thesis

3. Which of the following arguments by Hume claims that natural laws are not necessary (i.e., they are contingent)?

- a. The problem of induction
- b. The semantic argument
- c. The epistemic argument
- d. The metaphysical argument (Hume's "fork")

4. "Every time we have used induction in the past, it has worked, so it will probably work in the future."

- a. This is not a good justification for using induction because it is based on induction itself.
- b. This is not an attempt to justify the use of induction.
- c. This is a good justification for using induction, because it uses the rule of inductive inference, and therefore it is a strong argument
- d. This is a good justification for using induction, because it relies on many confirmations

5. What is problematic about the following statement: "An explanation of a phenomenon is scientific only if it specifies the causes of that phenomenon"?

- a. This is a redundant explanatory model, because there is already the deductive-nomological model
- b. This is a proposal for an explanatory model that is both too weak and too strong
- c. The concept of cause is no clearer than the concept of explanation
- d. None of the answers (except this one) is correct

6. We found a possible world in which the premises of an argument are false and the conclusion is also false. This indicates that:

- a. The argument is invalid
- b. The argument is valid in some worlds but not in all
- c. The argument may be invalid
- d. The argument is not sound

7. We are given a strong argument. Upon examining its assumptions, we find that they are all true. From this we can conclude that...

- a. The argument is sound
- b. The argument is deductive
- c. The argument is reliable
- d. The argument may not be reliable

8. Which of the following is not possible regarding a given scientific theory?

- a. That it is true
- b. To know that it is true
- c. That it is not true
- d. All answers are incorrect

9. According to the deductive-nomological model of explanation (Hempel),

- a. The height of the flagpole does not explain the length of its shadow, but the length of the shadow does explain the height of the flagpole

- b. The height of the flagpole explains the length of its shadow, and the length of the shadow explains the height of the flagpole
- c. The height of the flagpole does not explain the length of its shadow, and the length of the shadow does not explain the height of the flagpole
- d. The height of the flagpole explains the length of its shadow, but the length of the shadow does not explain the height of the flagpole

10. Based on observations, Galileo determined that the acceleration of a free falling body on Earth is 9.8 m/s^2 . From Newton's laws, we get that $a = GM/R^2$. If we plug-in the mass of the Earth M and its radius R into Newton's equation (and the gravitational constant G), we get that the acceleration is 9.8 m/s^2 . This indicates that:

- a. The logical directions in the two laws are opposite to one another
- b. Newton's law proves the truth of Galileo's law
- c. Newton's law can be formulated by induction from Galileo's observations
- d. Newton's law explains Galileo's law

11. Consider two good abductive arguments. In both arguments the premise is an observation and the conclusion is a theory. In such a situation, it is impossible to arrive at...

- a. A theory that fails to provide an explanation for the observation in the premise of the argument
- b. Two different theories that explain the same observations
- c. The same theory that explains different observations
- d. One false theory and one true theory

12. The Duhem-Quine thesis is a critique of the scientific method of refutation by deduction, because it shows that...

- a. Although there is no problem with deductive arguments, there is a fundamental problem with applying them in scientific practice to refute a theory or hypothesis.
- b. There is a problem with deductive arguments. They may lead to absurd results
- c. Although there is no problem with deductive arguments and no problem with their application, in practice scientists remain faithful to their hypotheses even when there are observations that contradict them
- d. The scientific method makes it too easy to disprove almost any hypothesis

13. The main problem with the (positivist) scientific method of confirmation is

- a. that it is difficult to understand
- b. that it may lead to absurd results
- c. that it cannot prove the truth of a theory
- d. that scientists are not familiar with it

14. Which of the following arguments is not valid?

- a. There are no married bachelors. Therefore, the president of Russia is Putin.
- b. The president of Russia is Putin. Therefore, the president of Russia is Putin.
- c. The president of France is Putin. Therefore, there are no married bachelors.
- d. There are married bachelors. Therefore, the president of France is Putin.

15. Does the scientific method allow an observation of a green sock today to generate the prediction that the first horse observed after 2050 will be blue?

- a. Yes. Using the description “Here is a Sorse that is blue” (“Sorse” = everything that was first observed before 2050 and is a sock, or was not first observed before 2050 and is a horse).

b. Yes. Using the description “here is a horse that is grue” (‘grue’ = anything first observed before 2050 that is green or not first observed before 2050 that is blue)

c. No.

d. Yes. Using the description “here is a non-blue object that is not a horse”

16. Why, according to the Armstrong-Dretske-Tully view, is the statement “Whenever Lucy wears green, the team wins” an accidental generalization and not a law of nature, even though it is true (assuming it is true)?

a. Because there is no connection between the property of wearing green and the property of winning the game.

b. Because the statement is not logically necessary.

c. Because the statement is too restrictive (it contains a proper name).

d. Because, according to Hume, there is no real difference between natural laws and accidental generalizations. Ultimately, they are just constant conjunctions.

17. The multiple realizations view challenges the possibility of mental explanations by physical ones. What is not true according to the multiple realizations position?

a. There may be a type of mental state that is not identical to a type of brain state, even though every instance of a mental state is identical to an instance of a brain state.

b. It may be the case that there are mental laws and physical laws, but it is not possible to translate mental laws into physical laws.

c. There may be an instance of a mental state that is not identical to an instance of a mental state.

d. There may be no bridging statements between mental terms and physical terms.

18. Which of the following is NOT true regarding biology in the context of the Unity of Science ideal (synchronic reduction)?

- a. The fact that biology has very few laws (some argue that there is only one law, which is natural selection) challenges the very possibility of explaining psychology and sociology through biology.
- b. The fact that both biology and psychology have a teleological aspect eliminates the need for bridge laws in order to explain psychology by biology.
- c. The theory of evolution removed a fundamental limitation regarding the reduction of biology to physics.
- d. In order for chemistry to be able to explain biology, bridge laws are required to translate biological terms into chemical terms.

19. Which of the following components of the scientific method is not criticized by Kuhn?

- a. The relationship between observation and theory
- b. The diachronic relationship between different theories
- b. The synchronic relationship between different theories
- d. None of the answers (except this one) is correct

20. Which of the following does not characterize a paradigm?

- a. It includes the criteria for evaluating a theory
- b. It consists of a network of terms of two types: theoretical terms and observational terms
- c. It dictates how we interpret observations
- d. It includes the meanings of the terms that appear in it

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Formula sheet

I. Rules of deductive inference

Premises	Direction of entailment	Conclusion	Rule of inference
1. If P then Q 2. P	$\rightarrow$	Q	MP
1. If P then Q 2. Not-Q	$\rightarrow$	Not-P	MT
1. If P then Q 2. If Q then R	$\rightarrow$	If P then R	HS
1. If P then Q	$\leftrightarrow$	If not-Q then not-P	Trans
1. If P then Q	$\leftrightarrow$	Not-P or Q	Impl
1. P or Q 2. not-P	$\rightarrow$	Q	DS
P	$\rightarrow$	P or Q	Add
P and Q	$\rightarrow$	P	Simp

II. Truth tables of logical connectives

p	q	p and q	p or q	if p then q	p if and only if q
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

II. Rules of inductive inference

Statistical Syllogism	Inductive Generalization	Combined Inductive Generalization	Analogical Inference
1. Most F's are G 2. x is F 3. x is G	1. Most F's examined are G 2. Representative amount examined 3. Most F's are G	1. Most F's examined are G 2. Representative amount examined 3. x is F 4. Therefore, x is G	1. Most of what's true for x is true for y 2. x is F 3. Therefore, y is F